

CS604

OPERATING SYSTEM SHORT NOTES

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Q1. Deadlock handling method

There are three method of deadlock handling

- Deadlock Prevention
- Deadlock Avoidance
- Allowing Deadlocks and Recovering

Q2. Deadlock Prevention Method

There are four necessary conditions

- Mutual Exclusion
- Hold and Wait
- No Preemption
- Circular wait

Q3. Difference between logical and physical address space

1. The basic difference between Logical and physical address is that Logical address is generated by CPU in perspective of a program whereas the physical address is a location that exists in the memory unit.
2. Logical Address Space is the set of all logical addresses generated by CPU for a program whereas the set of all physical address mapped to corresponding logical addresses is called Physical Address Space.
3. The logical address does not exist physically in the memory whereas physical address is a location in the memory that can be accessed physically
4. The logical address is generated by the CPU while the program is running whereas the physical address is computed by the Memory Management Unit (MMU).

Q4. Calculate physical address for the given logical address?

Logical address(16 bit)
IP= 0B10h

CS = D000h

Physical address(20 bit)

CS*24+IP = D0B10h

Q5. Define overlays?

Overlay is a technique to run a program that is bigger than the size of the physical memory by keeping only those instructions and data that are needed at any given time. Divide the program into modules in such a way that not all modules need to be in the memory at the same time.

Advantage

Reduce memory requirement

Reduce time requirement

Disadvantage

Overlap map must be specified by programmer

Programmer must know memory requirement

Overlapped module must be completely disjoint

Programmimg design of overlays structure is complex and not possible in all cases

Q6. Define Frames vs Pages?

Physical memory is broken down into fixed-sized blocks, called **frames**.

Logical memory is divided into blocks of the same size, called **pages**.

The size of a page is a **power of 2**.

Q7. Structure of the page table?

The following schemes allow efficient implementations of page tables

- Hierarchical/ Multilevel paging
- Hashed page table
- Inverted page table

Q8. What is segmentation?

A process is divided into Segments. The chunks that a program is divided into which are not necessarily all of the same sizes are called segments. Segmentation gives user's view of the process which paging does not give. Here the user's view is mapped to physical memory.

There are types of segmentation:

1. **Virtual memory segmentation**

Each process is divided into a number of segments, not all of which are resident at any one point in time.

2. **Simple segmentation**

Each process is divided into a number of segments, all of which are loaded into memory at run time, though not necessarily contiguously.

Advantages of Segmentation

- No Internal fragmentation.
- Segment Table consumes less space in comparison to Page table in paging.

Disadvantage of Segmentation

- As processes are loaded and removed from the memory, the free memory space is broken into little pieces, causing External fragmentation.

There are two registers in segmentation
Segment-table base register(STBR)
Segment-table length register(STLR)

Q9. What is virtual memory?

Virtual memory is the separation of user logical memory from physical memory. Virtual memory makes the task of programming easier because the programmer need not worry about the amount of physical memory.

Virtual memory is commonly implemented as demand paging. It can also be implemented in a segmentation system.

Q10. Demand Paging?

Demand paging is identical to the paging system with swapping. In demand paging, a page is delivered into the memory on demand i.e., only when a reference is made to a location on that page. Demand paging combines the feature of simple paging and implement virtual memory as it has a large virtual memory. Lazy swapper concept is implemented in demand paging in which a page is not swapped into the memory unless it is required.

Q11. Define Pre-paging?

A pure demand paging system is the large number of page faults that occur when a process is started. Pre-paging is an attempt to prevent this high level of initial paging. The strategy is to bring into memory at one time all the pages that will be needed.

Q12. Write the name of disk-scheduling algorithms?

The popular disk-scheduling algorithms are

- First-come-first-serve(FCFS)
- Shortest seek time first(SSTF)
- Scan
- Look
- Circular scan (C-scan)
- Circular look (C-Look)

Q13. Write the Type of bits associated with each entry of segment table?

The bits associated with each entry in the segment table , for the purpose of protection are.

Validation bit: if the validation bit is 0 , it indicates an illegal segment.

Read, write,execute bits.

Q14. Differentiate briefly between MFT and MVT?

1)MFT (Multiprogramming with a Fixed number of Tasks) is one of the old memory management techniques in which the memory is partitioned into fixed size partitions and each job is assigned to a partition.

2)MFT suffers with the problem of internal fragmentation.

1)MVT (Multiprogramming with a Variable number of Tasks) is the memory management technique in which each job gets just the amount of memory it needs. That is, the partitioning of memory is dynamic and changes as jobs enter and leave the system. MVT is a more "efficient" user of resources.

2)MVT suffers with external fragmentation.

Q13. What steps needed for page replacement?

1. Find the location of the desired page on the disk

2. Find a free frame

a) If there is a free frame use it.

b) If there is no free frame, use a page replacement algorithm to select a victim frame.

3. Read the desired page into the newly freed frame; change the page and frame tables.

4. Restart the user process

Q14. Write the names of three commonly used methods for file space allocation?

The three commonly used methods for file space allocation.

- 1) Contiguous allocation
- 2) Linked allocation
- 3) Indexed allocation

Q15. Write the name of correct system calls?

1	When a file is closed, the OS removes its entry in the open-file table.	close
2	A directory is searched for the appropriate entry and the current-file-position is set to a given value	Reposition (lseek)
3	A user may want to erase the contents of the file but keep its attributes.	Truncate (creat)
4	Search the directory for the named file, and then release the file space and erase the directory entry.	Delete(unlink)

Q16. The operating system monitors CPU utilization and, if CPU utilization is too low, the operating system increases the degree of multiprogramming by introducing one or more new processes to the system. This decreases the number of frames allocated to each process currently in the system, causing more page faults and further decreasing the CPU utilization. according to the given scenario Write the name of this technique?

Thrashing.

Q17. what is mounting? And its types of mounting?

We want to access any file firstly we have to add that file to the system that is mounting. Mounting operation can be done only by system administrator and hence it acts as a protection to the file system.

Types of mounts

- Remote mount
- Local mount

Q18. Difference between Remote mounts and Local mounts?

Remote mounts are done on a remote system on which data is transmitted over a telecommunication line.

Local mounts are mounts done on your local system.

Q19. Define file attributes?

Name: The symbolic file name is the only information kept in human-readable form.

Type: This information is needed for those systems that support different types.

Location: This location is a pointer to a device and to the location of the file on that device.

Size: The current size of the file (in bytes, words or blocks) and possibly the maximum allowed size are included in this attribute.

Protection: Access control information determines who can do reading , writing, etc. Owner

Time and date created: useful for security, protection and usage monitoring.

Time and date last updated: useful for security, protection and usage monitoring.

Read/write pointer value

Q20. Consider a process with its segment 15 having 5096 bytes. The process generates a logical address (15,3921). Is it a legal address? How many pages does the segment have? What page does the logical address refer to? Is it a legal address? Yes

How many pages does the segment have?

$$\text{ceiling}[5096/1024] = 5$$

What page does the logical address refers to?

$$\text{ceiling}[3921/1024] = 4 \text{ (i.e., page number 3)}$$

What are the value of d' and the physical address if page number 3 (i.e., the fourth page) is in frame 12? Here is how we compute these parameters, along with the graphical representation of the various parameters. Logical to physical address translation is shown in the figure on the next page.

$$d' = 3921 - 3*1K = 849$$

$$\text{Physical address} = 12*1K + 849 = 13137$$

Q21. calculate effective access time ?

$$T_{\text{mem}} = 100\text{nsec}$$

$$T_{\text{TTLB}} = 20 \text{ nsec}$$

Hit ratio is 80% $T_{\text{effective}} = 0.8 (20 + 100) + 0.2 (20 + 2 \times 100)$ nanoseconds = 140 nanoseconds

This means that with 80% chances of finding a page table entry in the TLB, the effective access time becomes 40% worse than memory access time without paging.

Q22. Let's consider the following example to explain how a deadlock avoidance algorithm works. There is a system with 12 tape drives and three processes?

Process	Max Need	Allocated	Available
P1	10	5	3
P2	4	2	5
P3	9	2	10

Q23. Define Classic Problems of Synchronization?

The three classic problems of synchronization are:

- Bounded-Buffer Problem
- Readers and Writers Problem
- Dining Philosophers Problem

Q24. Define types of semaphores?

There are two kinds of semaphores:

▪ **Counting semaphore** whose integer value can range over an unrestricted integer domain.

▪ **Binary semaphore** whose integer value cannot be > 1 ; can be simpler to implement.

Q25. Define semaphores and its types?

A semaphore S is an integer variable that, apart from initialization is accessible only through two standard atomic operations: wait and signal.

Q26. write the solution of Dining Philosophers Problem?

Allow at most four philosophers to be sitting simultaneously at the table.

Allow a philosopher to pick up her chopsticks only if both chopsticks are available (to do this she must pick them up in a critical section)

Use an asymmetric solution; that is, an odd philosopher picks up first her left chopstick, whereas an even philosopher picks up her right chopstick and then her left chopstick.

Q27. Assume a logical address space of 16 pages of 1024 words, each mapped into a physical memory of 32 frames. Here is how you calculate the various parameters related to paging.?

Solution:

No. of bits needed for p = ceiling $\lceil \log_2 16 \rceil$ bits = 4 bits

No. of bits needed for f = ceiling $\lceil \log_2 32 \rceil$ bits = 5 bits

No. of bits needed for d = ceiling $\lceil \log_2 2048 \rceil$ bits = 11 bits

Logical address size = $|p| + |d| = 4 + 11$ bits = 15 bits

Physical address size = $|f| + |d| = 5 + 11$ bits = 16 bits

Best Of Luck

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